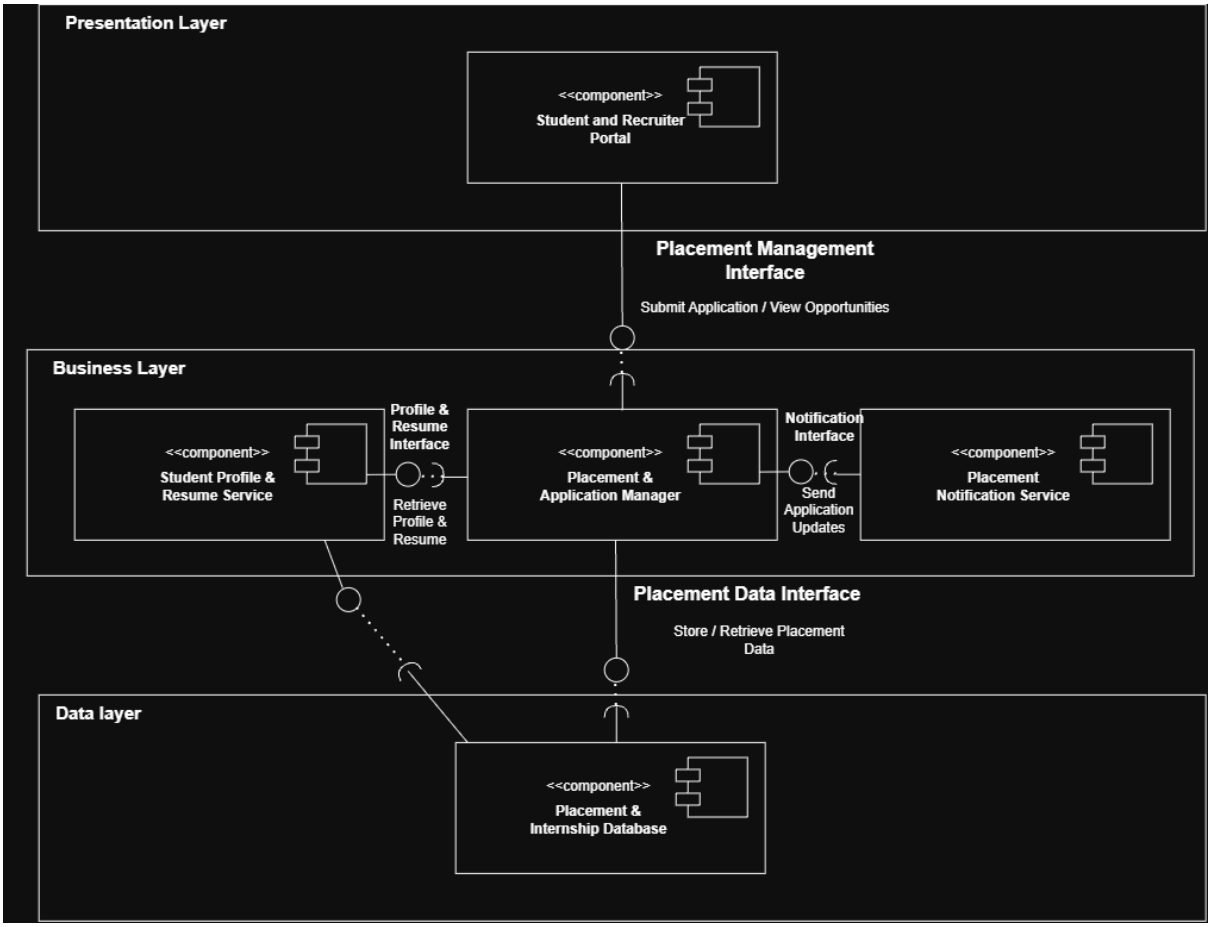
SE LAB 3

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Section G

1. Component Diagram



2.Written Justification – Campus Placement and Internship System

The Campus Placement and Internship System uses a Layered Architecture consisting of three main layers: the Presentation Layer, Business Layer, and Data Layer. This architectural style was selected because it separates the system into components with clearly defined responsibilities, making the system easier to develop, maintain, and modify.

The Presentation Layer contains the Student and Recruiter Portal component. This component provides the user interface through which students and recruiters can interact with the system. Users can view placement opportunities and submit applications through the Placement Management Interface.

The Business Layer contains the main application logic. The Student Profile and Resume Service manages student profiles and resumes. The Placement and Application Manager manages placement opportunities and student applications. It retrieves profile and resume information through the Profile and Resume Interface and sends application updates through the Notification Interface. The Placement Notification Service is responsible for sending notifications to users regarding application updates and placement-related information.

The Data Layer contains the Placement and Internship Database. The Placement and Application Manager communicates with this component through the Placement Data Interface to store and retrieve placement and internship information. The Student Profile and Resume Service also interacts with the database to store and retrieve profile and resume data.

The use of interfaces between components reduces direct dependency and provides clear communication paths. This improves modularity because individual components can be modified or replaced with minimal impact on other parts of the system. The layered architecture also improves maintainability and separation of concerns, as user interface functions, business logic, and data management are organised independently.

Overall, this architecture is suitable for the Campus Placement and Internship System because it supports clear component responsibilities, structured data flow, and easier future expansion.